

$$1) \lim_{x \rightarrow \infty} \frac{x^2 \cdot \cos(2x!)}{x^4 + 3} = ?$$

- a) 1 **b) 0** c) Limit mevcut değildir d) 1/3 e) 2

$$\lim_{x \rightarrow \infty} \underbrace{\frac{x^2}{x^4 + 3}}_0 \cdot \underbrace{\cos(2x!)}_{-1 \leq A \leq 1} = \underline{\underline{0}}$$

2) $f(x) = \sqrt{2 - \sqrt{3 - \sqrt{4 - x}}}$ fonksiyonunun tanım kümesi aşağıdakilerden hangisidir?

- a) (3, 4) b) (-5, 4] c) [3, 4] **d) [-5, 4]** e) Hiçbiri

$$\begin{aligned} 2 - \sqrt{3 - \sqrt{4 - x}} &\geq 0 \quad \vee \quad 3 - \sqrt{4 - x} \geq 0 \quad \vee \quad 4 - x \geq 0 \\ \downarrow & \qquad \qquad \qquad \downarrow & \qquad \qquad \qquad \downarrow \\ 2 &\geq \sqrt{3 - \sqrt{4 - x}} & 3 &\geq \sqrt{4 - x} & 4 &\geq x \\ \downarrow & \qquad \qquad \qquad \downarrow & \qquad \qquad \qquad \downarrow \\ 4 &\geq 3 - \sqrt{4 - x} & 9 &\geq 4 - x & & \\ \downarrow & \qquad \qquad \qquad \downarrow & \qquad \qquad \qquad \downarrow \\ \sqrt{4 - x} &\geq -1 & x &\geq -5 & & \\ \downarrow & \qquad \qquad \qquad \downarrow & \qquad \qquad \qquad \downarrow \\ \sqrt{4 - x} &\text{ tanımlı} & & & & \\ \text{iken} &\text{ yeri} & & & & \\ 4 - x \geq 0 &\rightarrow x \leq 4 \text{ iken} & & & & \\ &\text{doğru} & & & & \end{aligned}$$

$\rightarrow$ **[-5, 4]**

③ $\lim_{x \rightarrow 0} \frac{x^{2020} \sin 4x}{(\sin x)^{2021}} ?$

a) 0 b) 2 **c) 4** d) 2020 e) hiçbir

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{x^{2020} \cdot \sin 4x}{(\sin x)^{2021}} &= \lim_{x \rightarrow 0} \frac{x^{2020}}{(\sin x)^{2020}} \cdot \frac{\sin 4x}{\sin x} \\ &= \lim_{x \rightarrow 0} \underbrace{\left(\frac{x}{\sin x} \right)^{2020}}_1 \cdot \frac{\sin 4x}{4x} \cdot \underbrace{\frac{x}{\sin x}}_1 \cdot 4 \\ &= 4 \end{aligned}$$

④ $\lim_{x \rightarrow 1} \frac{|x-1|+x}{|x-4|} ?$

a) limit mevcut değildir b) 1/2 **c) 1/3** d) 1/4 e) 1

$$\begin{aligned} \lim_{x \rightarrow 1^+} \frac{x-1+x}{-(x-4)} &= +\frac{1}{3} & \lim_{x \rightarrow 1^-} \frac{1-x+x}{-(x-4)} &= \frac{1}{3} \\ &\nearrow \quad \nwarrow & & \\ &\text{Limit } \frac{1}{3} & & \end{aligned}$$

5) $\lim_{x \rightarrow -\infty} \sqrt{4x^2 + 2x} + 2x = ?$

- a) 0
- b) ∞
- c) $-\infty$
- d) $\frac{1}{2}$

e) $-\frac{1}{2}$

$$\begin{aligned} \lim_{x \rightarrow -\infty} \frac{\sqrt{4x^2 + 2x} + 2x}{(\sqrt{4x^2 + 2x} - 2x)} &= \lim_{x \rightarrow -\infty} \frac{\cancel{4x^2} + 2x - \cancel{4x^2}}{\sqrt{4x^2 + 2x} - 2x} = \lim_{x \rightarrow -\infty} \frac{2x}{\frac{1}{-x} \left(\sqrt{4 + \frac{2}{x}} - 2x \right)} \\ &= \lim_{x \rightarrow -\infty} \frac{2x}{-x \left[\sqrt{4 + \frac{2}{x}} + 2 \right]} = \frac{2}{-4} = -\frac{1}{2} \end{aligned}$$

6)

$\lim_{x \rightarrow 0} \frac{(\sin x)^{100}}{x^{99} \sin 2x}$ limitinin sonucu aşağıdakilerden hangisidir?

- a) 2
- b) $\frac{1}{2}$
- c) 0
- d) ∞

e) Limit mevcut değildir.

$$\lim_{x \rightarrow 0} \frac{(\sin x)^{100}}{x^{99} \sin 2x} = \lim_{x \rightarrow 0} \frac{(\sin x)^{99}}{x^{99}} \cdot \frac{\sin x}{\sin 2x} = \frac{1}{2}$$

7) $\lim_{x \rightarrow \infty} x^2 \cdot \left(1 - \cos \frac{2}{x}\right) = ?$ a) 0 b) 1 c) 2 d) $\frac{1}{2}$

$$\lim_{x \rightarrow \infty} \frac{x^2 \cdot \left(1 - \cos \frac{2}{x}\right) \cdot \left(1 + \cos \frac{2}{x}\right)}{1 + \cos \frac{2}{x}} = \lim_{x \rightarrow \infty} \frac{x^2 \cdot \left(1 - \cos^2 \frac{2}{x}\right)}{1 + \cos \frac{2}{x}}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{\sin^2 \frac{2}{x}}{1 - \cos^2 \frac{2}{x}}}{\frac{1}{x^2}} \cdot \frac{1}{1 + \cos \frac{2}{x}}$$

$$= \lim_{x \rightarrow \infty} \left(\frac{\frac{\sin \frac{2}{x}}{\frac{2}{x}}}{1} \right)^2 \cdot 4 \cdot \frac{1}{\frac{1 + \cos \frac{2}{x}}{2}} = \underline{\underline{2}}$$

8) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin |x| \cdot \tan x} = ?$ a) 0 b) 1 c) $\frac{1}{2}$ d) Limit mevcut değil

$$\lim_{x \rightarrow 0^+} \frac{1 - \cos x}{\sin x \cdot \tan x} = \lim_{x \rightarrow 0^+} \frac{(1 - \cancel{\cos x}) \cdot \cos x}{\sin^2 x} = \lim_{x \rightarrow 0} \frac{\cos x}{\frac{1 - \cos^2 x}{(1 - \cancel{\cos x})(1 + \cos x)}} = \frac{1}{2}$$

$$\lim_{x \rightarrow 0^-} \frac{1 - \cos x}{-\sin x \cdot \tan x} = \lim_{x \rightarrow 0^-} - \frac{(1 - \cancel{\cos x}) \cdot \cos x}{\sin^2 x} = \lim_{x \rightarrow 0^-} - \frac{\cos x}{1 + \cos x} = -\frac{1}{2}$$

$$\lim_{x \rightarrow 0^+} \frac{1 - \cos x}{\sin |x| \cdot \tan x} \neq \lim_{x \rightarrow 0^-} \frac{1 - \cos x}{\sin |x| \cdot \tan x} \Rightarrow \text{limit mevcut değil}$$

9) $f(x) = \frac{\sqrt{12x-3}-x}{x^3-1}$ fonksiyonunun tanım kümesi?

① $12x-3-x \geq 0$ olması $\rightarrow$ $2x-3 \geq x \rightarrow \boxed{x \geq 3}$
veya
 $2x-3 \leq -x \rightarrow \boxed{x \leq 1}$

② $x^3-1 \neq 0$ olması $\Rightarrow \boxed{x \neq 1}$ olması

① ve ② den $D(f) = (-\infty, 1) \cup [3, \infty)$

10) $\lim_{x \rightarrow \infty} \frac{x - \sqrt{1+x^2}}{x - \sqrt{x}} = ?$

a) 1 b) 2 c) 0 d) ∞ e) $-\infty$

$\lim_{x \rightarrow \infty} \frac{x(1 - \sqrt{1 + \frac{1}{x^2}})}{x(1 - \frac{1}{\sqrt{x}})} = \boxed{0}$

11) $\lim_{x \rightarrow 0} \frac{1 - \sqrt{1 - \sin 2x}}{\tan x} = ?$ a) 0 b) 1 c) 2 d) ∞

$\lim_{x \rightarrow 0} \frac{1 - \sqrt{1 - \sin 2x}}{\tan x} = \lim_{x \rightarrow 0} \frac{(1 - \sqrt{1 - \sin 2x})(1 + \sqrt{1 - \sin 2x})}{\tan x \cdot (1 + \sqrt{1 - \sin 2x})}$

$= \lim_{x \rightarrow 0} \frac{x - x + \sin 2x}{\tan x \cdot (1 + \sqrt{1 - \sin 2x})} = \lim_{x \rightarrow 0} \frac{2 \cdot \cancel{\sin x} \cdot \frac{1}{\cancel{\cos x}}}{\frac{\sin x}{\cos x} \cdot (1 + \sqrt{1 - \sin 2x})}$
 $= \frac{2}{2} = \underline{\underline{1}}$

12) $f(x) = \frac{\sqrt{|2x+5|}-3}{x^2+1}$ Tanım Kümesi?

a) $[-1, \infty) \cup (-\infty, -4]$

b) $(-1, \infty) \cup (-\infty, -4)$

c) $(4, \infty) \cup (-\infty, 1)$

d) $[4, \infty) \cup (-\infty, 1]$

$|2x+5|-3 \geq 0$ olmalı.

$|2x+5| \geq 3 \rightarrow 2x+5 \geq 3 \rightarrow x \geq -1 \rightarrow$
 $\rightarrow 2x+5 \leq -3 \rightarrow x \leq -4 \rightarrow D(f) = (-\infty, -4] \cup [-1, \infty)$

13) $f(x) = \frac{1}{\sqrt{|x|-x}}$ Tanım kümesi?

a) $(0, \infty)$

b) $(-\infty, 0)$

c) $(1, \infty)$

d) $(-\infty, 1)$

$|x|-x > 0$ olmalı $\Rightarrow |x| > x \Rightarrow x < 0 \Rightarrow D(f) = (-\infty, 0)$

14) $\lim_{x \rightarrow \infty} (\sqrt{x^2+x} - \sqrt{x^2-x}) = ?$

a) 0

b) 1

c) 2

d) $\frac{1}{2}$

e) -1

$\lim_{x \rightarrow \infty} \frac{(\sqrt{x^2+x} - \sqrt{x^2-x}) \cdot (\sqrt{x^2+x} + \sqrt{x^2-x})}{\sqrt{x^2+x} + \sqrt{x^2-x}} = \lim_{x \rightarrow \infty} \frac{x^2+x - x^2+x}{\sqrt{x^2+x} + \sqrt{x^2-x}}$

$= \lim_{x \rightarrow \infty} \frac{2x}{\sqrt{x^2+x} + \sqrt{x^2-x}} = \lim_{x \rightarrow \infty} \frac{2}{\frac{1}{x}(\sqrt{1+\frac{1}{x}} + \sqrt{1-\frac{1}{x}})} = \frac{2}{2} = 1$

⑮ $\lim_{x \rightarrow 0} \frac{\sin x^2}{\sqrt{1+x^2}-1} = ?$ a) 0 b) 1 c) 2 d) $\frac{1}{2}$ e) -1

$$\lim_{x \rightarrow 0} \frac{\sin x^2}{\sqrt{1+x^2}-1} = \lim_{x \rightarrow 0} \frac{\sin x^2}{1+x^2-1} \cdot (\sqrt{1+x^2}+1) = \lim_{x \rightarrow 0} \frac{\sin x^2}{\frac{x^2}{1}} \cdot \frac{(\sqrt{x^2+1}+1)}{2} = 2$$

⑯ $\lim_{x \rightarrow \infty} x^2 \cdot \sin \frac{1}{x} = ?$ a) 0 b) 1 c) 2 d) ∞

I. 401

$$\lim_{x \rightarrow \infty} x \cdot \frac{\sin \frac{1}{x}}{\frac{1}{x}} = \infty$$

⑰ $\lim_{x \rightarrow \infty} \sqrt{x} \cdot \sin \frac{1}{x} = ?$ a) 0 b) 1 c) 2 d) ∞

$$\lim_{x \rightarrow \infty} \frac{\sin \frac{1}{x}}{\frac{1}{\sqrt{x}}} \cdot \frac{\frac{1}{\sqrt{x}}}{\frac{1}{\sqrt{x}}} = \lim_{x \rightarrow \infty} \frac{\sin \frac{1}{x}}{\frac{1}{x}} \cdot \frac{\frac{1}{\sqrt{x}}}{\frac{1}{x}} = 0$$

⑱ $\lim_{x \rightarrow 0^-} \frac{\sqrt{1-\cos 2x}}{\sin 2x} = ?$ a) 0 b) 1 c) $\frac{1}{\sqrt{2}}$ d) $-\frac{1}{\sqrt{2}}$

$$\lim_{x \rightarrow 0^-} \frac{\sqrt{1-\cos 2x}}{\sin 2x} \cdot \frac{\sqrt{1+\cos 2x}}{\sqrt{1+\cos 2x}} = \lim_{x \rightarrow 0^-} \frac{\sqrt{1-\cos^2 2x}}{\sin 2x} \cdot \frac{1}{\sqrt{1+\cos 2x}}$$

$$= \lim_{x \rightarrow 0^-} \frac{\frac{-\sin 2x}{|\sin 2x|}}{\sin 2x} \cdot \frac{1}{\sqrt{1+\cos 2x}} = \lim_{x \rightarrow 0^-} \frac{-\sin 2x}{\underbrace{\sin 2x}_{-1}} \cdot \frac{1}{\underbrace{\sqrt{1+\cos 2x}}_{\frac{1}{\sqrt{2}}}} = -\frac{1}{\sqrt{2}}$$

19) $\lim_{x \rightarrow 1} \frac{\sqrt[3]{x} - 1}{\sqrt{x} - 1} = ?$ a) 0 b) 1 c) $\frac{3}{4}$ d) $\frac{4}{3}$

$x = t^12$ olsun. $x \rightarrow 1 \Rightarrow t \rightarrow 1$

$$\lim_{t \rightarrow 1} \frac{t^4 - 1}{t^3 - 1} = \lim_{t \rightarrow 1} \frac{\cancel{t^3} - 1)(t^2 + 1)}{\cancel{t^3} (t^2 + t + 1)} = \frac{2 \cdot 2}{3} = \frac{4}{3}$$

20) $\lim_{x \rightarrow 3^+} \frac{\sin \sqrt{x-3}}{\sqrt{\sin(x-3)}} = ?$ a) 0 b) 1 c) -1 d) ∞

$$\lim_{x \rightarrow 3^+} \underbrace{\frac{\sin \sqrt{x-3}}{\sqrt{x-3}}}_{1} \cdot \underbrace{\frac{\sqrt{x-3}}{\sqrt{\sin(x-3)}}}_{\sqrt{\frac{x-3}{\sin(x-3)}}} = \boxed{1}$$

21) $\lim_{x \rightarrow \infty} \left(x - \frac{x^2}{\sqrt{x^2+x}} \right) = ?$ a) 0 b) 1 c) $\frac{1}{2}$ d) ∞

$\rightarrow \infty - \infty$ belirsizliği

$$\lim_{x \rightarrow \infty} \frac{x \cdot \sqrt{x^2+x} - x^2}{\sqrt{x^2+x}} = \lim_{x \rightarrow \infty} \frac{x^2(x^2+x) - x^4}{\sqrt{x^2+x} \cdot (x\sqrt{x^2+x} + x^2)}$$

$$\frac{(x\sqrt{x^2+x} + x^2)}{\frac{1 \cdot 1}{x} \cdot \sqrt{1+\frac{1}{x}} \cdot \frac{1 \cdot 1}{x} \cdot \sqrt{1+\frac{1}{x}}}$$

$$= \lim_{x \rightarrow \infty} \frac{\cancel{x^3}^1}{\cancel{x^3} \cdot \sqrt{1+\frac{1}{x}} \cdot \left(\sqrt{1+\frac{1}{x}} + 1 \right)} = \frac{1}{2}$$

$$(22) \quad \lim_{x \rightarrow 0} \frac{1 - \sqrt{1 + \sin x}}{1 - \sqrt{1 - x}} = ?$$

$$(1 + \sqrt{1 + \sin x}) \cdot (1 + \sqrt{1 - x})$$

$$\lim_{x \rightarrow 0} \frac{(\cancel{x} - \cancel{x} - \sin x) \cdot (1 + \sqrt{1 - x})}{(1 + \sqrt{1 + \sin x}) (\cancel{x} - \cancel{x} + x)} = \lim_{x \rightarrow 0} - \frac{\sin x}{x} \cdot \frac{1 + \sqrt{1 - x}}{1 + \sqrt{1 + \sin x}} = (-1)$$

$\frac{2}{2} = 1$

$$1) \lim_{x \rightarrow \pi} \frac{\sin 2x}{x^2 - \pi x} = ?$$

a) 0 **b) $\frac{2}{\pi}$** c) ∞ d) 1

I. yol
 $x - \pi = t$ olsun. $x \rightarrow \pi$ ise $t \rightarrow 0$ olur.

$$\lim_{x \rightarrow \pi} \frac{\sin 2x}{x(x - \pi)} = \lim_{t \rightarrow 0} \frac{\sin(2\pi + 2t)}{(\pi + t) \cdot t} = \lim_{t \rightarrow 0} \underbrace{\frac{\sin 2t}{2t}}_1 \cdot \frac{2}{\pi + t} = \frac{2}{\pi}$$

II. yol
 $\sin 2x = -\sin(2\pi - 2x)$

$$\lim_{x \rightarrow \pi} \frac{\sin 2x}{x(x - \pi)} = \lim_{x \rightarrow \pi} \frac{\sin(2\pi - 2x)}{x \cdot \underbrace{(x - \pi)}_{-(\pi - x)}} = \lim_{x \rightarrow \pi} \underbrace{\frac{\sin(2\pi - 2x)}{2(\pi - x)}}_{\frac{\sin 0}{0} \rightarrow 1} \cdot \underbrace{\frac{2}{x}}_{\frac{2}{\pi}} = \frac{2}{\pi}$$

2) $\lim_{x \rightarrow -\infty} (x^2 + x\sqrt{x^2 - 1}) = ?$ a) 0 b) 2 **c) $\frac{1}{2}$** d) $-\frac{1}{2}$ e) ∞
 $\infty - \infty \rightarrow$ Eşlenik çarpım

$$\lim_{x \rightarrow -\infty} \frac{(x^2 + x\sqrt{x^2 - 1})}{(x^2 - x\sqrt{x^2 - 1})} = \lim_{x \rightarrow -\infty} \frac{x^4 - x^2 \cdot (x^2 - 1)}{x^2 - x \cdot \sqrt{x^2 - 1}} = \lim_{x \rightarrow -\infty} \frac{x^2}{x^2 - x\sqrt{x^2 - 1}}$$

$$= \lim_{x \rightarrow -\infty} \frac{x^2}{x^2 - x \sqrt{x^2(1 - \frac{1}{x^2})}} \quad |x| = -x$$

$$= \lim_{x \rightarrow -\infty} \frac{\cancel{x^2}}{\cancel{x^2} \left[1 + \sqrt{1 - \frac{1}{\cancel{x^2}}} \right]} = \frac{1}{2}$$

3) Aşağıdaki limitlerden hangilerinin sonucu doğrudur?

I) $\lim_{x \rightarrow 1} (x-1) \cdot \sin \frac{1}{x-1} = 1$ II) $\lim_{x \rightarrow 2} \frac{\sin(x-2)}{x^2-4} = \frac{1}{4}$ III) $\lim_{x \rightarrow 0} \frac{\sin^2 2x}{x^2} = 2$ IV) $\lim_{x \rightarrow 0} \frac{1-\cos x}{x^2} = 0$

a) Yalnız I **b) Yalnız II** c) I ve II d) I, III ve IV e) I, II, III ve IV

$$\lim_{x \rightarrow 1} \underbrace{(x-1)}_0 \cdot \underbrace{\sin \frac{1}{x-1}}_{-1 \leq A \leq 1} = 0 \text{ dir (I Yanlıştır)}$$

$$\lim_{x \rightarrow 2} \frac{\sin(x-2)}{x^2-4} = \lim_{x \rightarrow 2} \frac{\sin(x-2)}{\frac{x-2}{1}} \cdot \frac{1}{\frac{x+2}{\frac{1}{4}}} = \frac{1}{4} \text{ (II doğru)}$$

$$\lim_{x \rightarrow 0} \frac{\sin^2 2x}{x^2} = \lim_{x \rightarrow 0} \frac{\sin 2x}{\frac{x}{2}} \cdot \frac{\sin 2x}{\frac{x}{2}} = 4 \text{ (III yanlıştır)}$$

$$\lim_{x \rightarrow 0} \frac{1-\cos x}{x^2} = \lim_{x \rightarrow 0} \frac{\sin^2 x}{x^2} \cdot \frac{1}{1+\cos x} = \frac{1}{2} \text{ (IV yanlıştır)}$$
$$\frac{(\frac{\sin x}{x})^2}{1} \cdot \frac{1}{\frac{1}{2}}$$